

NLP Final Project Proposal

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Motivation

The abilities of LLMs to solve math problems have been already been studied and improved to such extent that it is now considered to be an accelerator of academic math [1], [2]. Those achievements have been reached by training the models explicitly on math problem datasets such as GSM8k, and even the use of different tokens for math inputs [3]. There have been critics and thorough research questioning whether LMs reached a generalized understanding of math, or is it just hidden overfitting to the available math problems [4], [2].

In the work of Brown et al [5] models of various sizes are tested on explicit arithmetic problems of 2 to 3 digits, and show very little success in models smaller than 100B parameters, even with few-shot context.

To help complete the picture, we would like to test the success of those models on problems that are single digit, but implicit in phrasing. The challenge in those problems will not be the complex pattern recognition of 2 and 3 digit problems, but rather a linguistic and logical problem of inferring the mathematical operation out of natural language.

Textual arithmetic problems are a simple and well structured type of logical questions ("I ate one of my five apples, now I only have [blank]"). Answering them correctly is an evidence of deep logical abilities, even if the problem is mathematically simple.

Project Description

We are interested in the logical abilities that emerge naturally from modeling a language. Namely, we would like to answer the question - Does LMs trained on general text corpus reach any type of logical understanding? Specifically, we would like to explore the abilities of the models to infer the arithmetic question from natural text.

We are looking to find, out of a set of several models, the size and architecture of a model, and prompt structure, that shows success in such logical inference. Then we would like to outline their limitations across a few dimensions of complexity:

1. Number ranges - if it infers $1+1=2$ from the text could it infer $7+8=15$? What is the limit? Is it aligned with the frequency of the numbers in the training corpus?
2. Mathematical operation - addition, subtraction, multiplication, division
3. Implicit phrasing - "Two plus two makes [blank]" Vs "I have 2 kids, together with my wife we make a family of [blank]".
4. In context learning - Could presenting the model a few example questions help it answer correctly?

We thus would like to comprise our own small evaluation set of textual simple math problems, using code to extrapolate a set of templates to many examples. Evaluating the success could be done naively by requiring exact match to the correct answer, or in softer ways, like looking at the predicted probability of the correct answer.

Furthermore, we consider testing the connection between answering explicit arithmetic question and answering textual problems, for example, does fine tuning the model on explicit arithmetic questions improves its success on textual problems?

References

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